

EPISODE 5 KNOWLEDGE MINING LAB

Azure AI Search, Document Intelligence, and Content Understanding for the High-End Restaurant AI Automation solution

Audience: AI-102 practice builders | Scope: Episode 5 retrieval, ingestion, extraction, and analyzer workflows

This lab package extends the restaurant solution with a focused knowledge-mining layer. It gives you a practical path for testing retrieval quality, document extraction, and analyzer-driven multimodal review without having to redesign the earlier agentic or multilingual application slices.

What Is Included

Capability	Episode 5 Surface
Search	POST /search/query with keyword, vector, hybrid, and semantic-style retrieval
Operations	GET /search/ingest-status for a quick readiness and corpus check
Receipts	POST /document/receipt for line items, totals, and merchant metadata
Layout	POST /document/layout for paragraphs, tables, and page-level structure
Contracts	POST /document/contract for private event extraction and review routing
Analyzers	POST /content-understanding/analyze for menu, contract, audio, video, and invoice analyzer patterns

Suggested Lab Flow

1. Keep `MOCK_MODE=true` and validate the new endpoints locally before connecting live Azure services.
2. Run `scripts/build_search_index.py` to write the richer index schema and, when configured, create the Azure AI Search index.

3. Run scripts/ingest_documents.py to chunk restaurant source files, generate embeddings, and upload enriched search records.
4. Validate retrieval with questions about private dining minimums, vegan tasting availability, and shellfish references.
5. Upload a receipt, invoice, or event contract to confirm extraction quality and review-threshold behavior.
6. Test the content-understanding analyzers with representative PDF, audio, and video inputs.

Retrieval Defaults

Setting	Default	Usage Note
Query mode	hybrid	Balances lexical matching with vector-style grounding.
Result count	5	Keeps answers concise while still supporting citations.
Semantic config	restaurant-semantic	Used when semantic or hybrid ranking is enabled.
Vector size	3072	Matches the embedding shape used by the search ingestion flow.
Common filters	menu / policy / pairings / vegan	Use document_type, menu_section, and allergen_tag to narrow recall.

Why This Matters for AI-102

- Azure AI Search index design, chunking strategy, and citation-friendly retrieval outputs.
- Grounding prompts with explicit sources instead of relying on unsupported model memory.
- Document Intelligence extraction for structured and semi-structured restaurant operations documents.
- Separation of deterministic extraction logic from higher-level generative orchestration.
- Operational concerns such as ingest status, review thresholds, and metadata-driven filtering.